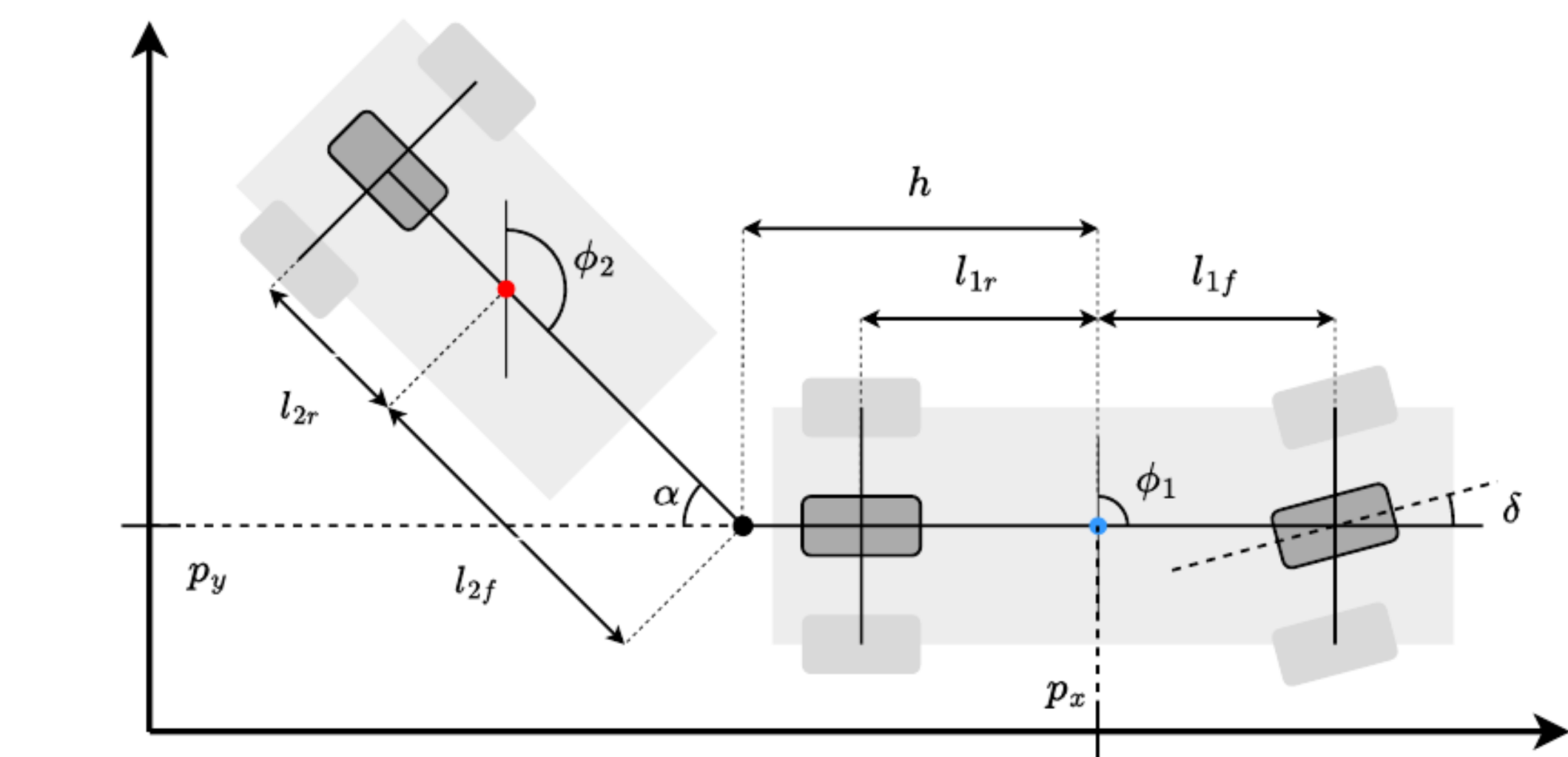


Learning Dynamics for High-Speed Tractor-Trailer Control

Aaron Chen* · David Shen* · Taekyung Kim* · Kaleb Ben Naveed · Dimitra Panagou

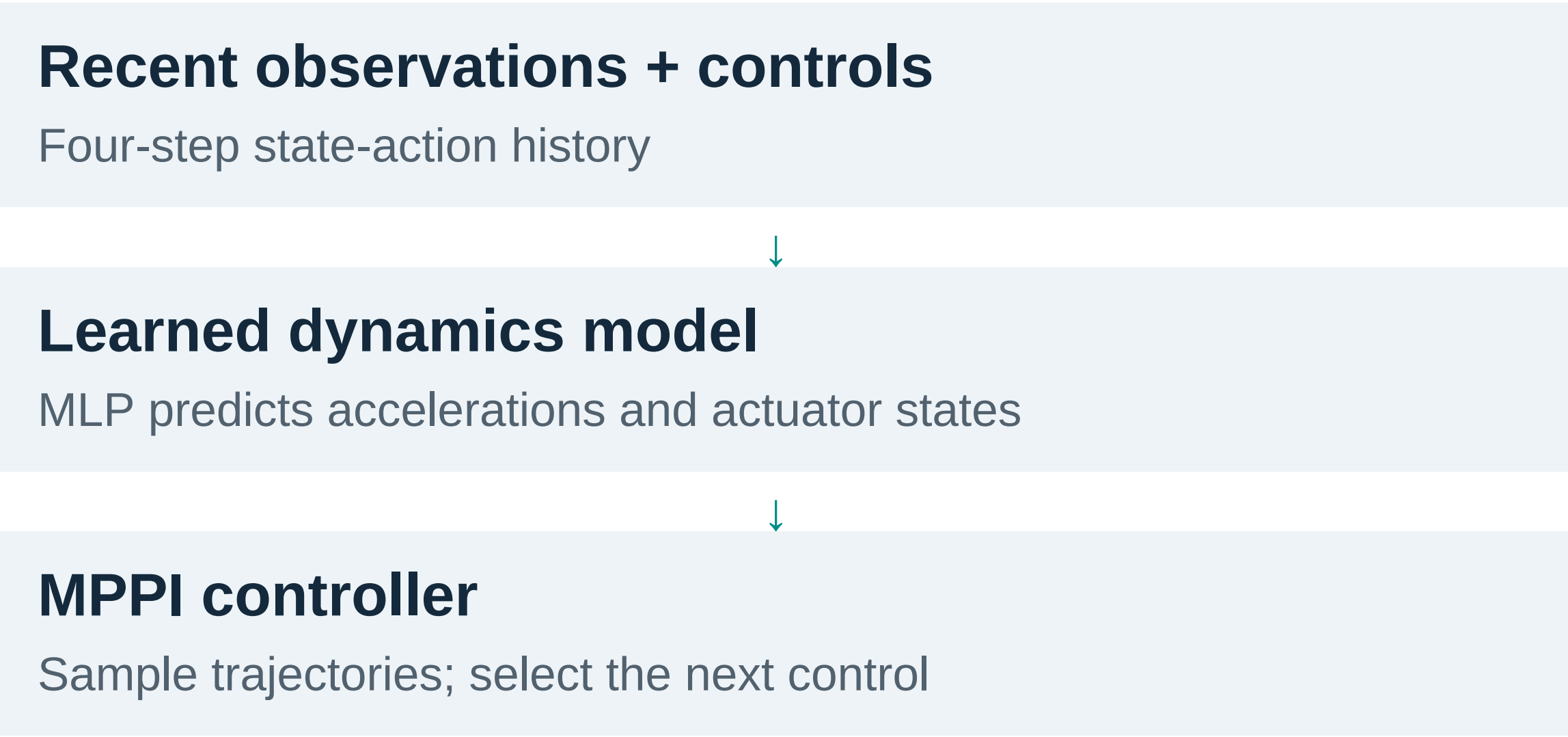
01 Why learn the dynamics?



Articulation makes reverse motion difficult.

- Reverse motion can amplify hitch-angle errors and lead to jackknifing.
- Friction, suspension, and other hidden dynamics affect high-speed behavior.
- Model mismatch degrades the trajectories predicted by MPC.

02 History → prediction → control

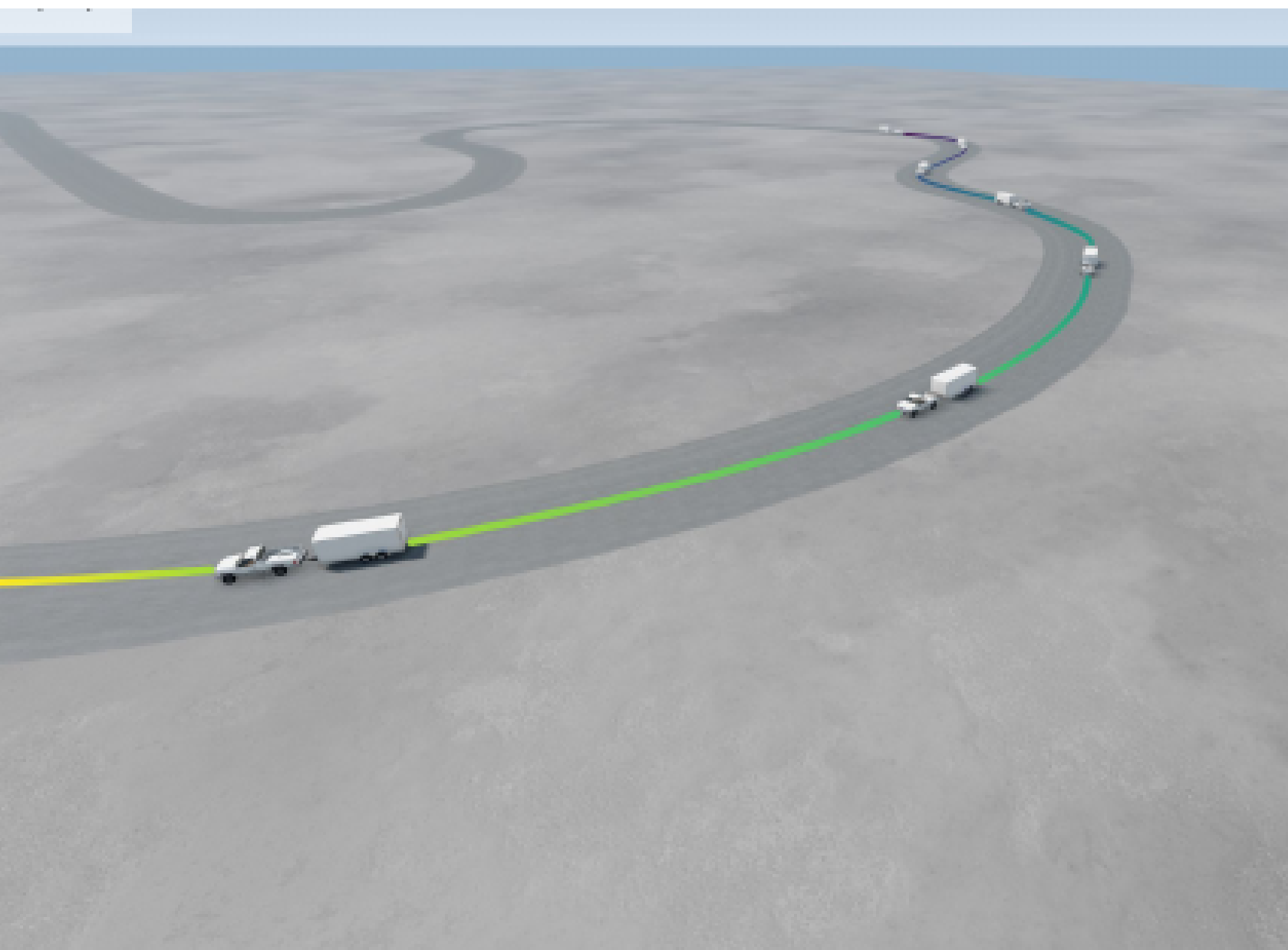


Training: varied friction and target speeds, with additional data near handling limits.

Analytic baseline: reduced-order tractor-trailer dynamics with Fiala tire forces.

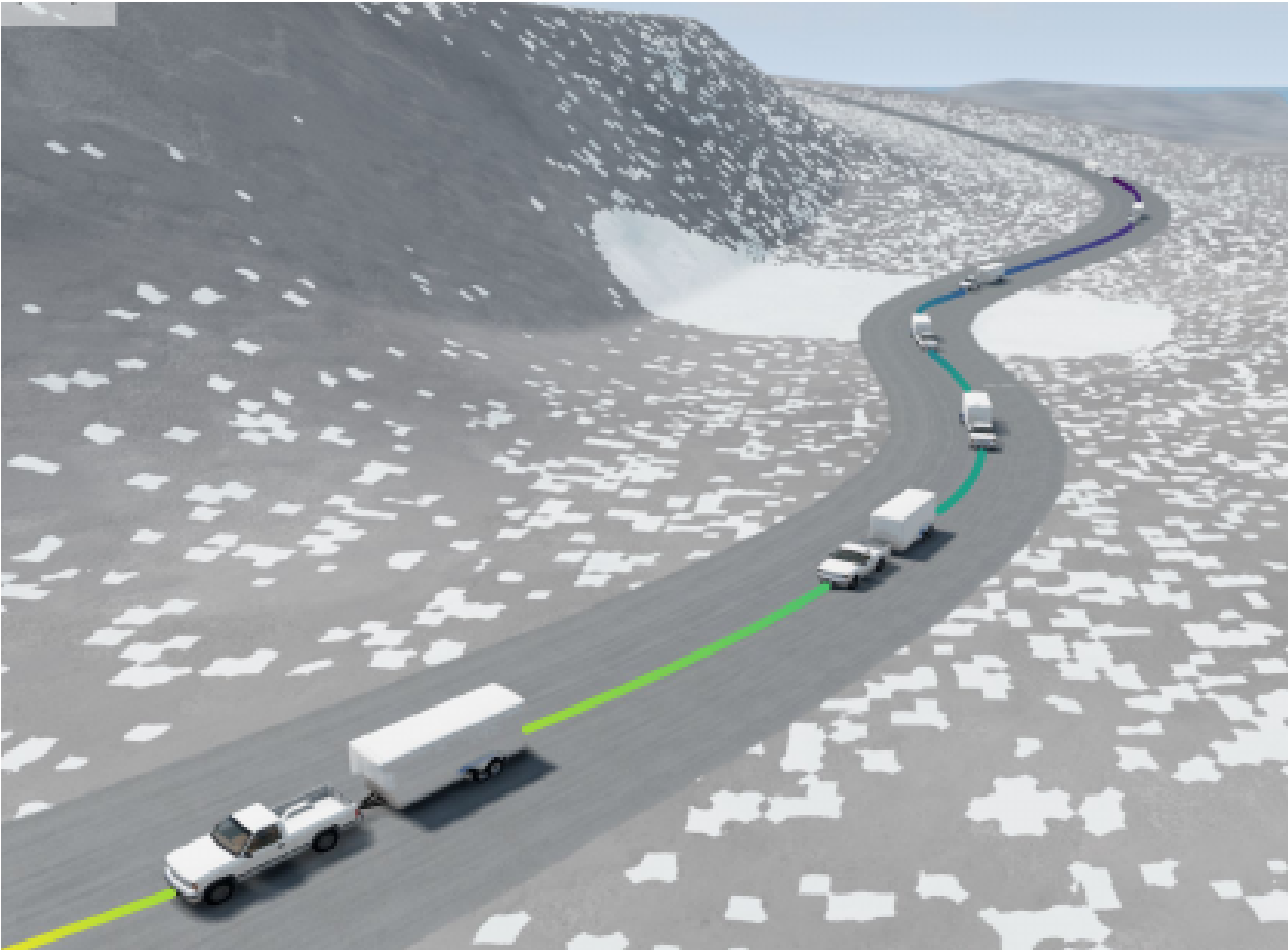
03 High-speed driving in simulation

REVERSE / ASPHALT

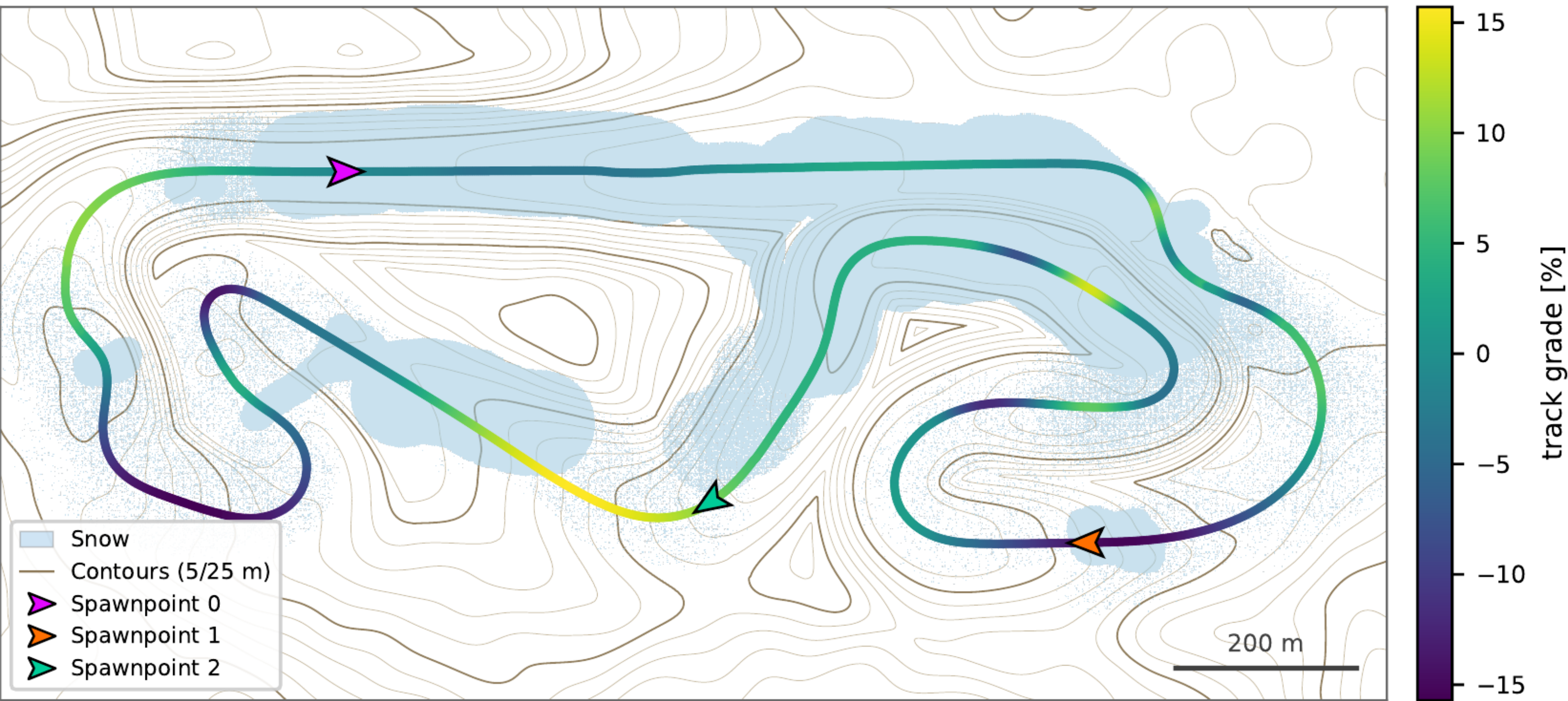


Representative trajectories with learned-dynamics MPPI. Colored paths show vehicle motion over time.

REVERSE / SNOW + SLOPE



04 Four environments. Unseen terrain.



ASPHALT / SNOW / SLOPE / SNOW + SLOPE

Target velocities: -30 and -50 km/h reverse; +80 km/h forward.
Protocol: nine runs per configuration in BeamNG.tech.
Comparisons: analytic-prior MPPI, LQR-PID, nonlinear MPC, MPCC.

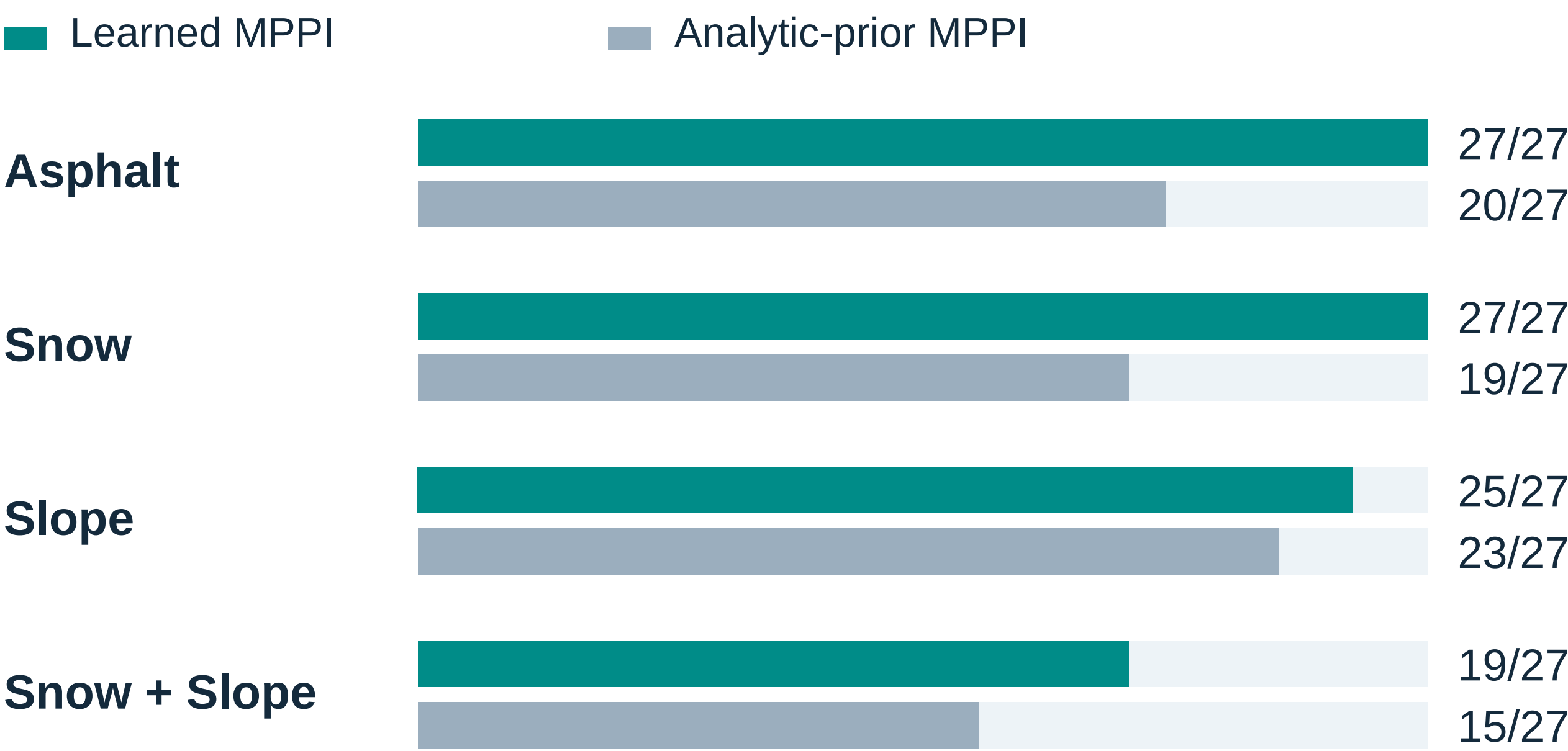
05 More reliable closed-loop control

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configurations with closer mean-speed tracking and lower RMS hitch angle than analytic-prior MPPI.

Runs reaching the 500-step limit

All three velocity setpoints pooled · 27 runs per environment



54/54 runs completed on flat Asphalt and Snow. Combined snow and slope still cause reverse-driving stalls.

06 Better multi-step prediction

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comparisons with the lowest reported open-loop RMSE.

- Lowest position RMSE in all four reverse-driving environments.
- Lowest hitch-angle RMSE in all four forward-driving environments.
- Some analytic-prior predictions remain more accurate; gains are not universal.

Scope: simulation only. Slope and banking were absent from training; payload variation and hardware tests remain future work.

Learned dynamics improve control under changing road conditions.

Transfer to tested unseen sloped terrain, with reverse-driving limits under combined snow and slope.